Pitch Angle Correction for DIC Measurements

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1. Introduction

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2. Yaw Correction

2.1. Problem Statement

2.2. Solution Approach

2.3. Results

3. Blade Pitch Angle Correction

3.1. Problem Statement

The second to last step in the currently employed DIC pipeline (see Appendix A), embodied in the *align_coordinate_system.py* script, relies on a series of linear transformations to eliminate the rigid body rotation of the wind turbine rotor. This process is then followed by exporting the location of a point across time for each of the identified AoIs on the blade, values which are then finally used to quantify their in-plane, out-of-plane and torsional deformation.

The flow diagram shown in Fig. 3.1 helps visualize the sequence of transformations carried out on the *.out* files within the script. These alignment transformations are carried out individually for each recorded frame on the complete point cloud (i.e. all valid points on all three blades), virtually eliminating both the rotor rotation and the changes in wind turbine yaw throughout the measurement.

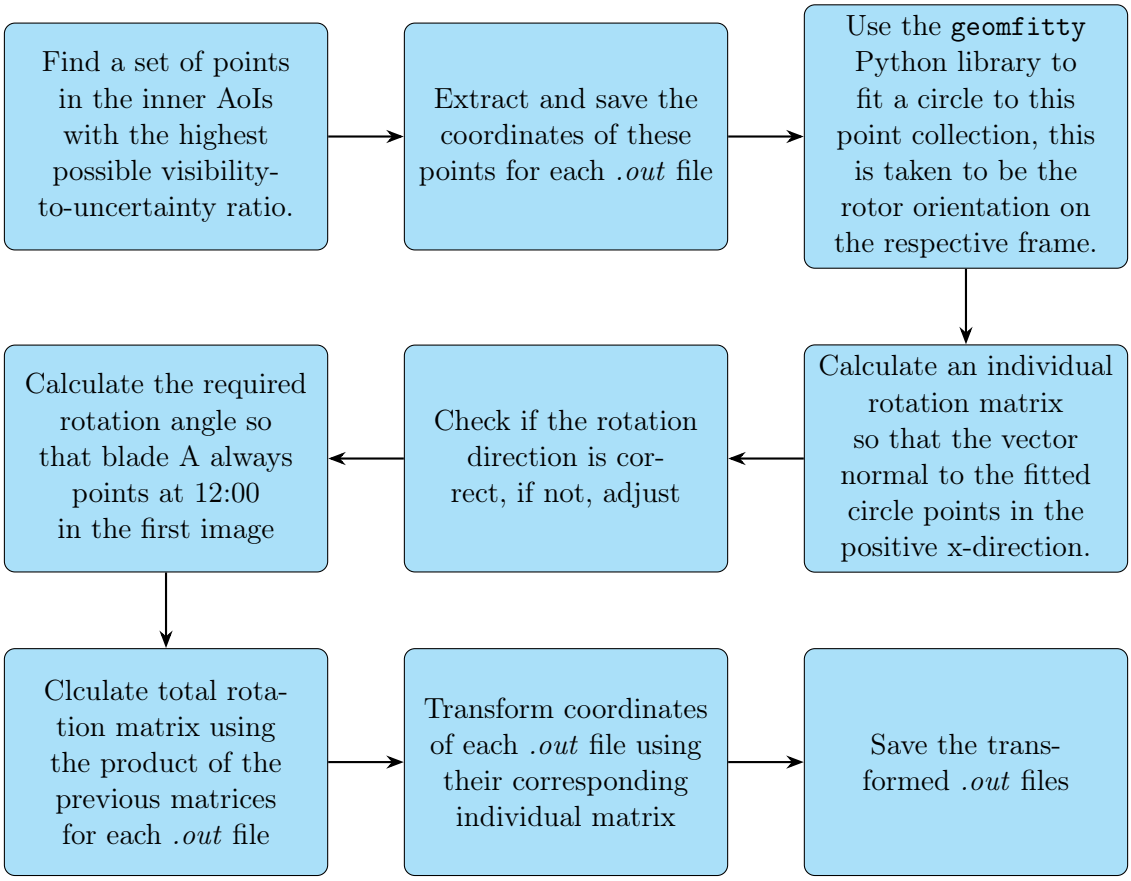


Figure 3.1.: Flow diagram of the transformations carried out within the original *align_coordinate_system.py* script.

Since the DIC measurement are carried out on a pitch-controlled turbine, each blade can also vary its pitch angle independently during operation. byproduct of applying global transformations to all points is the inability to remove the effects of these individual pitch rotations out of the points on each blade before calculating their deformation values. This manifests itself

as the presence of clearly blade pitch-dependent artifacts on the final deformation time-series. This effect is specially clear in the in-plane deformation values; a clear example of this can be seen in Fig. 3.2 using data from the *Messung 24* measurement.

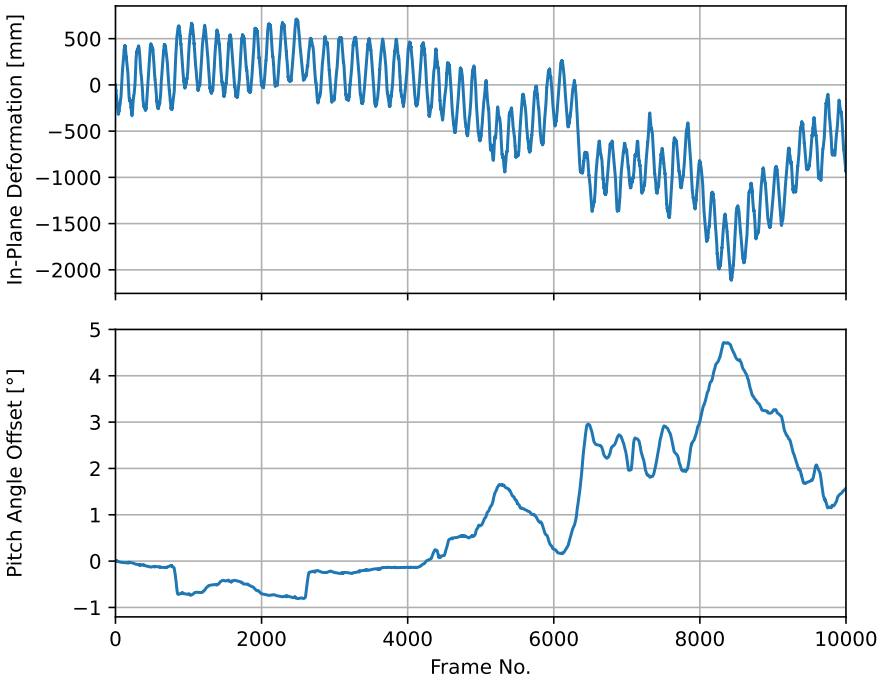


Figure 3.2.: In-plane deformation of blade A during the first 10,000 frames of the *Messung 24* measurement plotted on top of the pitch angle offset.

3.2. Correction Approach

Part of the dateset obtained from each field measurement *.mat* file containing a time series of the blade pitch angle value. In a nutshell, the correction process makes uses these values to virtually "un-pitch" all three blades individually in each frame. These new coordinate values are then stored in as a new set of *.out* files that are then used as base for the rest of the transformations carried out in the script.

It is assumed that all points in a given blade rotate about a fixed pitch axis throughout the measurement. The origin and orientation of this axis is approximated by fitting a cylinder to the innermost AoI of each blade; once all three pitch axes have been determined, the points corresponding to each blade are corrected using a sequence of lineal transformations in the following order:

1. Bring the blade axis origin to the global origin.
2. Rotate the blade points so that its pitch axis is aligned with the global z-axis.
3. Rotate the blade points about the corresponding pitch angle offset relative to the first frame (i.e. $\Delta\theta_i = \theta_i - \theta_0$) in the negative direction to "unpitch" the blade.
4. Apply the inverse rotation to bring the blade axis back to its original orientation.
5. Apply the inverse offset to bring back the blade origin to its original location.

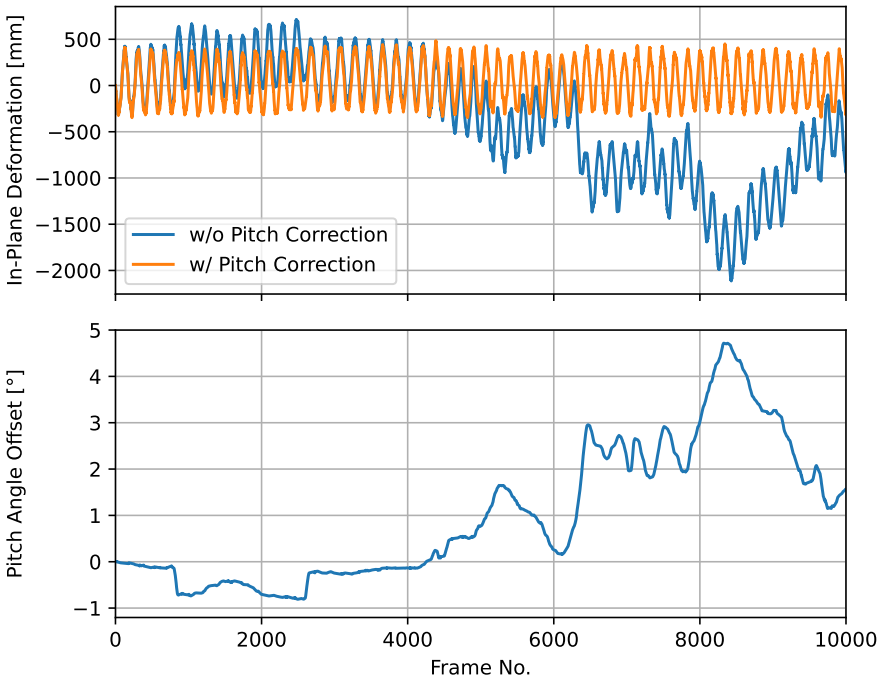


Figure 3.3.: In-plane deformation time-series for the third AoI in blade A with and without pitch correction.

Appendix B helps visualize the complete correction process with one a single blade (note that an exaggerated pitch value of 45 °C has been used there to aid in visualization). Once all three blades have been corrected for the current frame, a new *.out* file for that frame is written and exported for use in the remaining of the script.

3.3. Results & Validation

A direct comparison of the resulting in-plane deformation values with and without correction can be seen in Fig. 3.3. When compared to the original signal, it is pretty evident how the orange line, corresponding to the corrected values, has been decoupled from the changes in the blade pitch, oscillating with a uniform amplitude about a consistent centre value throughout the measurement.

Although these preliminary results looked qualitatively promising, since the dataset used during the development of the script emanated from a real-world measurement it was not possible to quantitatively evaluate how good the corrected results were with respect to a known baseline. With this in mind, a digital twin was used as a baseline .

The blender model used the pitch angle time series from measurement 24 as an input for pitch, the corrected in-plane and out-of-plane signals for its deformation. The frames were generated and processes using the pipeline shown in Appendix A. The resulting in-plane deformation curves can be seen in Fig. 3.4

The absolute error as a function of the deflection is shown in Fig. 3.5. In order to determine the measurement uncertainty, the deformation values are separated into 10 equally spaced bins. The error bars show the 95 % confidence interval for the mean as a measurement of uncertainty [1]. Table 3.1 summarizes the error measurements for both the uncorrected and corrected signals.

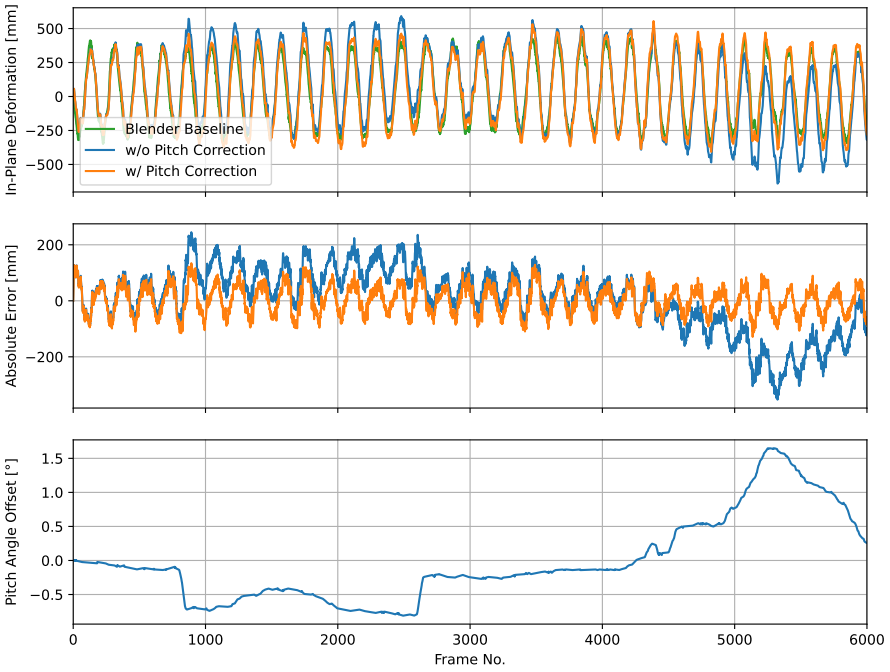


Figure 3.4.: Baseline, corrected, and uncorrected values for the synthetic measurement.

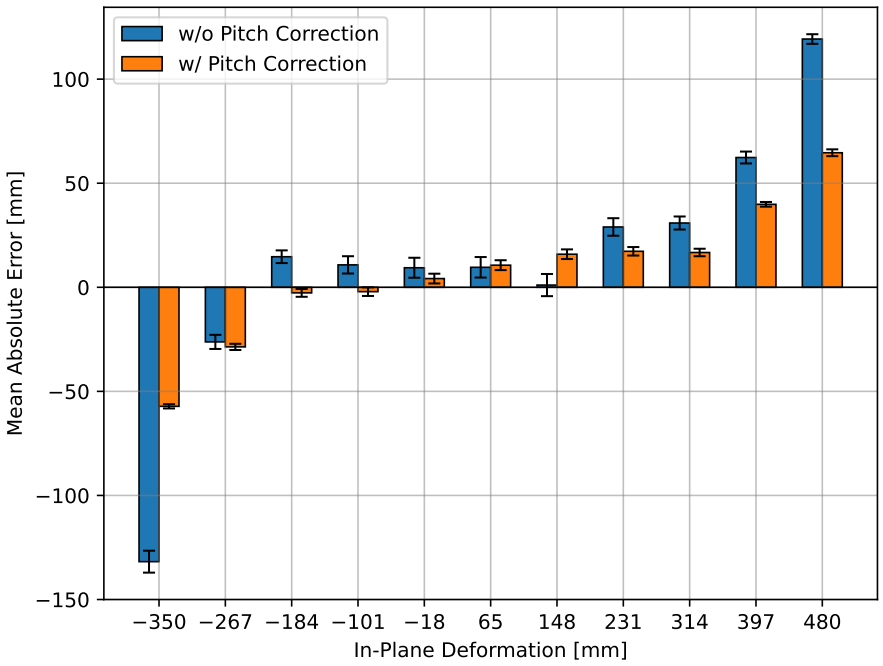


Figure 3.5.: Average error as a function of the in-plane deformation grouped in 10 equally spaced and sized bins.

Time Series	MAE	Max. Absolute Error	SEM (95 %)
w/o Pitch Correction	91.2 mm	353.0 mm	3.29 mm
w/ Pitch Correction	43.6 mm	127.4 mm	1.9 mm

Table 3.1.: Summary of error measurements for the corrected and uncorrected in-plane deformation signals.

References

[1] Jan Winstroth. “Optische Erfassung der Rotorblattverformungen von Windenergieanlagen der Multi-Megawatt-Klasse durch digitale Bildkorrelation”. Dissertation. Gottfried Wilhelm Leibniz Universität Hannover, Fakultät für Maschinenbau, 2020.

[2] Jan Winstroth. *Leitfaden: How to DIC a wind turbine*. Leitfaden. Arbeitsgruppe/Abteilung: TFD – Gruppe Turbine. Änderungsdatum: 04.05.2018 by Stephanie Lehnhoff. Leibniz Universität Hannover, Institut für Turbomaschinen und Fluid-Dynamik (TFD), Aug. 26, 2016.

A. WEA DIC Pipeline

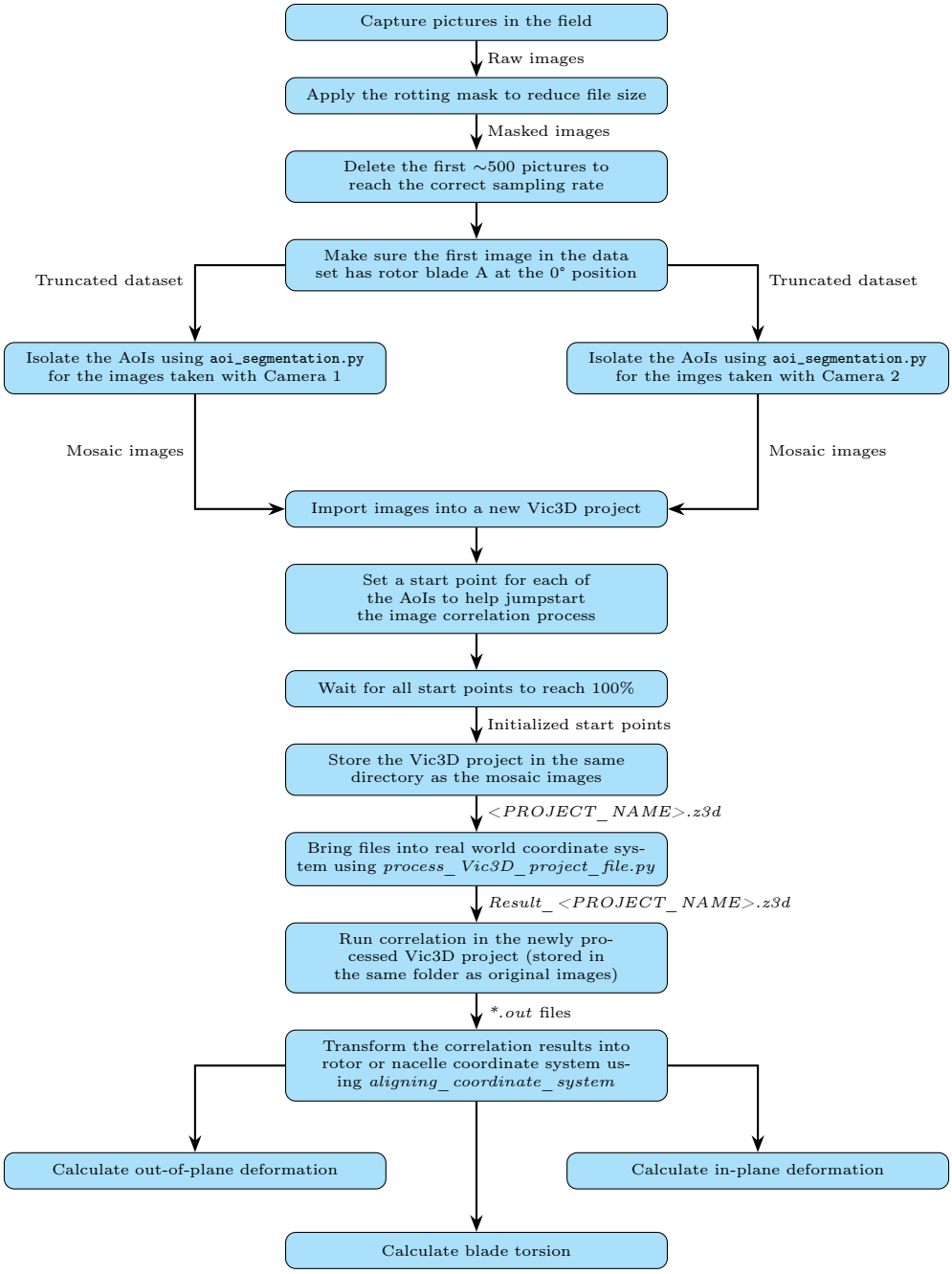


Figure A.1.: Complete processing pipeline for WEA DIC data acquisition and evaluation as described in [2].

B. Pitch Correction of a Single Blade

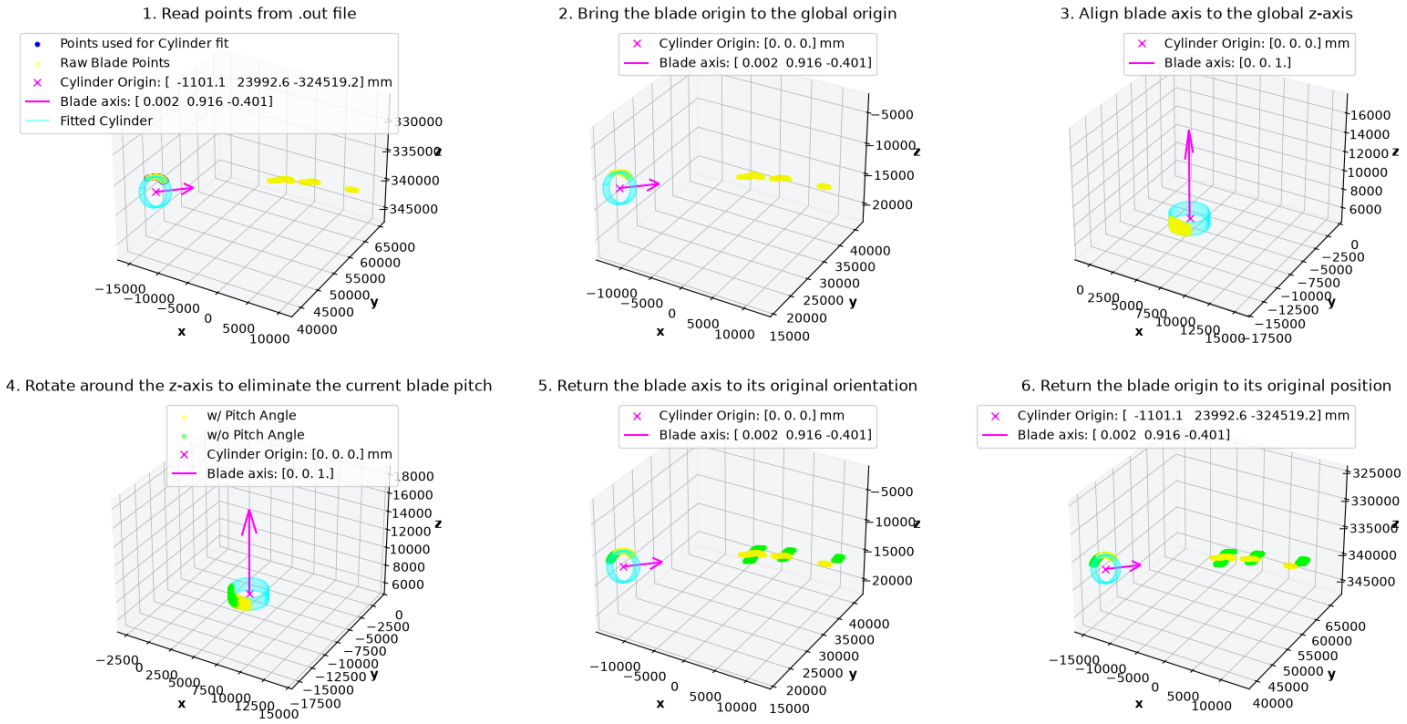


Figure B.1.: Pitch correction process exemplified with a single blade and a exagerated 45° pitch value.